

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think>

- The patient is a poultry worker with community-acquired pneumonia , treated with cephalexin , which is an antibiotic . This suggests a bacterial etiology , most likely Streptococcus pneumoniae .

- Streptococcus pneumoniae is known to cause community-acquired pneumonia and is sensitive to penicillin , which cephalexin is a member of .

- The immune response to Streptococcus pneumoniae involves both innate and adaptive immunity .

- Bacterial pathogens like Streptococcus pneumoniae are recognized by TLR4 on the surface of macrophages , not TLR5 .

- The immune response to Streptococcus pneumoniae is primarily Th1-mediated due to the production of pro-inflammatory cytokines .

- Peptidoglycan is a component of the bacterial cell wall but is not the major antigen for Streptococcus pneumoniae .

- Encapsulation is a mechanism used by some bacteria to evade the immune response , but Streptococcus pneumoniae does not have a capsule .

Therefore , the best answer is B , as Streptococcus pneumoniae primarily induces a Th1-cell response .

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<answer>B. It primarily induces the Th1-cell response.</answer>